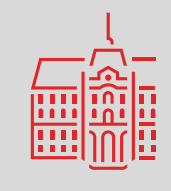


PIU: Proximity-guided Identity Unlearning in ID-Conditioned Diffusion Models

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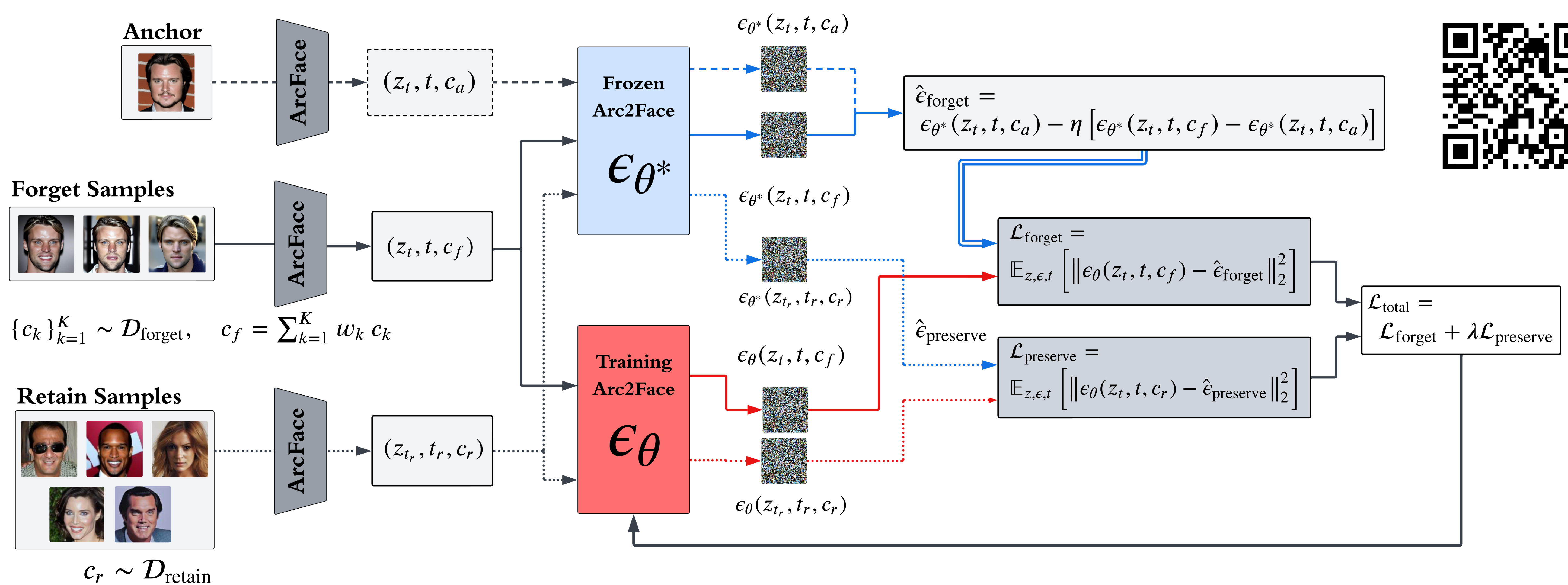


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Overview:

- **Privacy concerns:** Synthetic face generation raises privacy risks.
- **Right-to-be-forgotten:** Generative models lack identity unlearning mechanisms.
- **PIU framework:** Identity unlearning in ID-conditioned diffusion models.
- **ID unlearning:** Geometric identity replacement via proximity-based anchors.



Methodology:

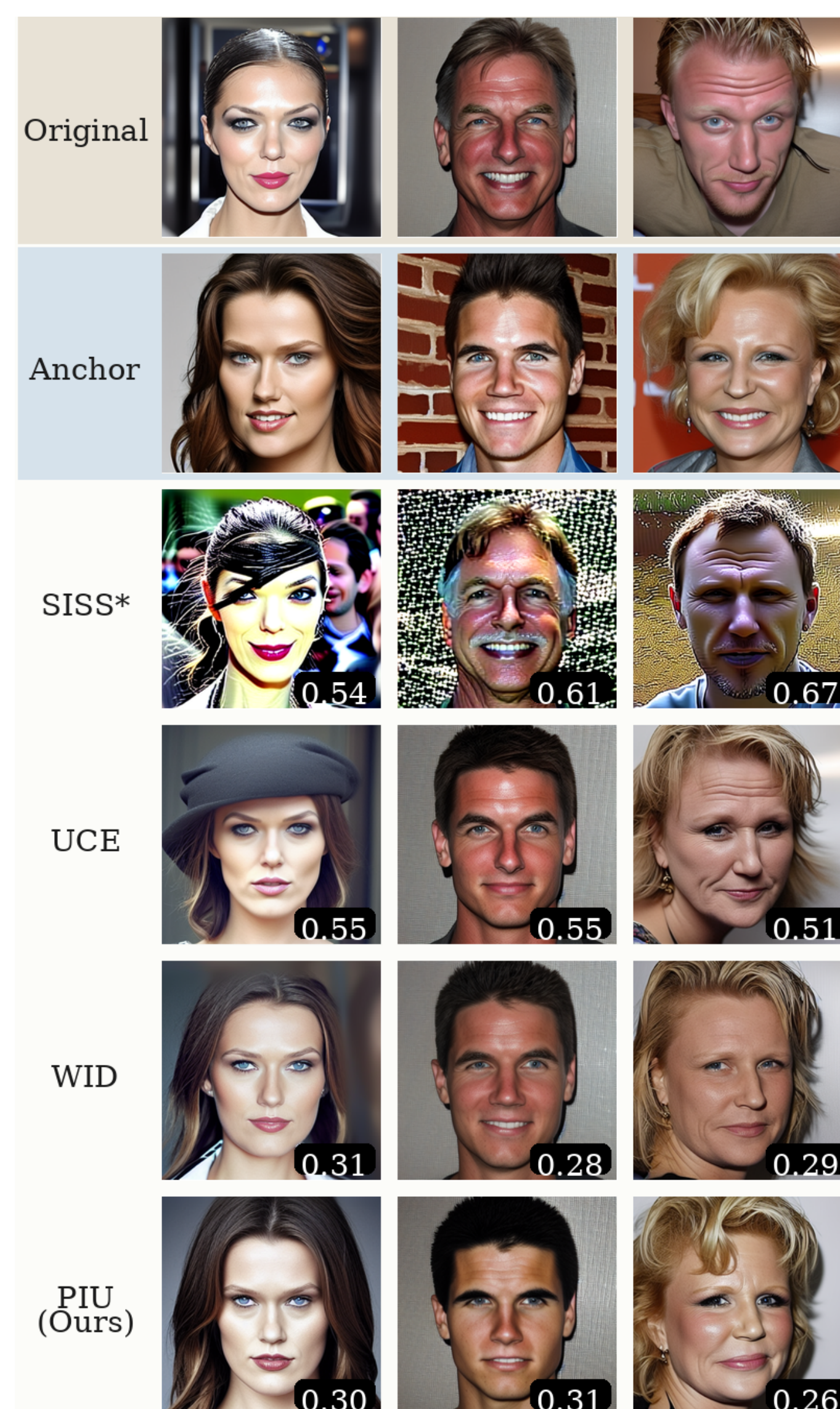
- **Forget loss:** Steers targets toward a selected anchor.
- **Preservation loss:** Preserves other non-target identities.
- **Anchors:** Selected via ArcFace cosine similarity.
- **Surgical fine-tuning:** Updates 4.29% of total parameters, targeting identity-sensitive cross-attention layers.

Experiments:

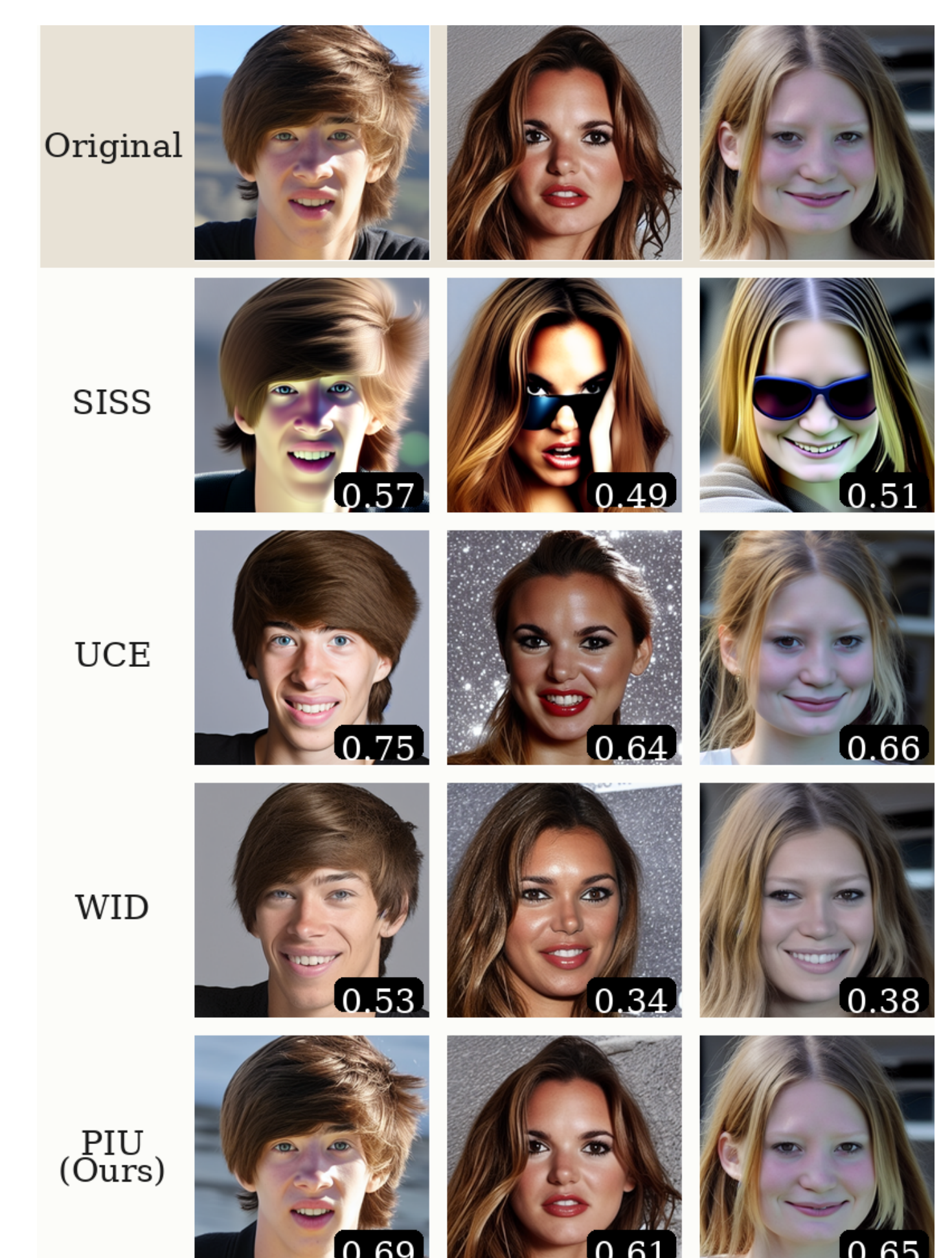
- **Base Model:** Arc2Face (ID-conditioned diffusion model).
- **Dataset:** CelebA-HQ (50 unlearning identities).
- **Methods:** SISS (data), UCE (concept), and WID (identity).
- **ID metrics:** ISM (ID erasure) & SRK (removal and keep).
- **Quality:** KD (kernel distance) & eDIFFIQA (face utility).

Model	ISM		SRK $\uparrow$	Δ KD		eDIFFIQA	
	Forget $\downarrow$	Retain $\uparrow$		Forget $\downarrow$	Retain $\downarrow$	Forget $\uparrow$	Retain $\uparrow$
Arc2Face	0.78	0.75	0.99	—	—	0.76	0.76
SISS	0.59	0.58	0.98	5.23	8.14	0.65	0.64
UCE	0.59	0.70	1.00	3.69	1.26	0.75	0.75
WID	0.32	0.44	49.28	4.35	3.36	0.76	0.75
PIU (Ours)	0.31	0.72	88.96	8.12	0.39	0.75	0.75

- **Results:** PIU delivers state-of-the-art identity unlearning (best forget ISM & SRK) with minimal disruption to retained identities (retain ISM & SRK) and quality (KD & eDIFFIQA).



Unlearned & Retained Identities



Takeaways:

- **Superior unlearning:** Steering targets toward suitable anchors ensures cleaner ID erasure than other methods.
- **Minimal collateral damage:** Preservation loss & local updates prevent catastrophic forgetting.
- **Practical privacy:** PIU provides a post-hoc solution for *Right to be Forgotten* compliance in generative models.